

École Pour l'Informatique et les Techniques Avancées – EPITA

Masters program

Course: Introduction to Network protocols and Architecture (INPA)

INPA Course Outline

Course schedule (tentative)

Date	No.	Topics	Duration (in hours)
(check Zeus platform for the date & time)	1	Primer, TCP/IP stack, TCP/IP stack operations, Class exercises	3
(check Zeus platform for the date & time)	2	IP address allocation & routing, Subnetting, ARP, Class exercises	3
(check Zeus platform for the date & time)	3	DHCP, Packet handling, Routing Protocols (RIP, OSPF), Class exercises	3
(check Zeus platform for the date & time)	4	TCP, UDP, ICMP, ON, Class exercise	3
(check Zeus platform for the date & time)	5	Internet layer protocols (IGP vs EGP, BGP), ACLs, Class exercises	3
(check Zeus platform for the date & time)	6	Application Layer protocols/Security (HTTP/S), Class exercise	3
(check Zeus platform for the date & time)	7	Application Layer protocols/Security (SMTP + PGP), DNS, Class exercises	3
(check Zeus platform for the date & time)	8	NAT, Class exercise, Review/Practice-session	3
Total			24 hours
(check Zeus platform for the date & time)	EXAM		1

GRADING criteria:

Class attendance and responsiveness:	10%
Class assignments (practical work):	40%
Final evaluation (Quiz & Exam):	50%

Lecture 4 Outline

- ▶ **Transmission Control Protocol (TCP)**
 - Implementation
 - Header
 - Versions
- ▶ **User Datagram Protocol (UDP)**
 - Implementation
 - Header
- ▶ **Internet Control Message Protocol (ICMP)**
 - Implementation
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 - Use cases
- ▶ **Optical Network (ON)**
 - ONUs and Types of setups
 - Why do we need ONs
- ▶ **Class exercise 10**

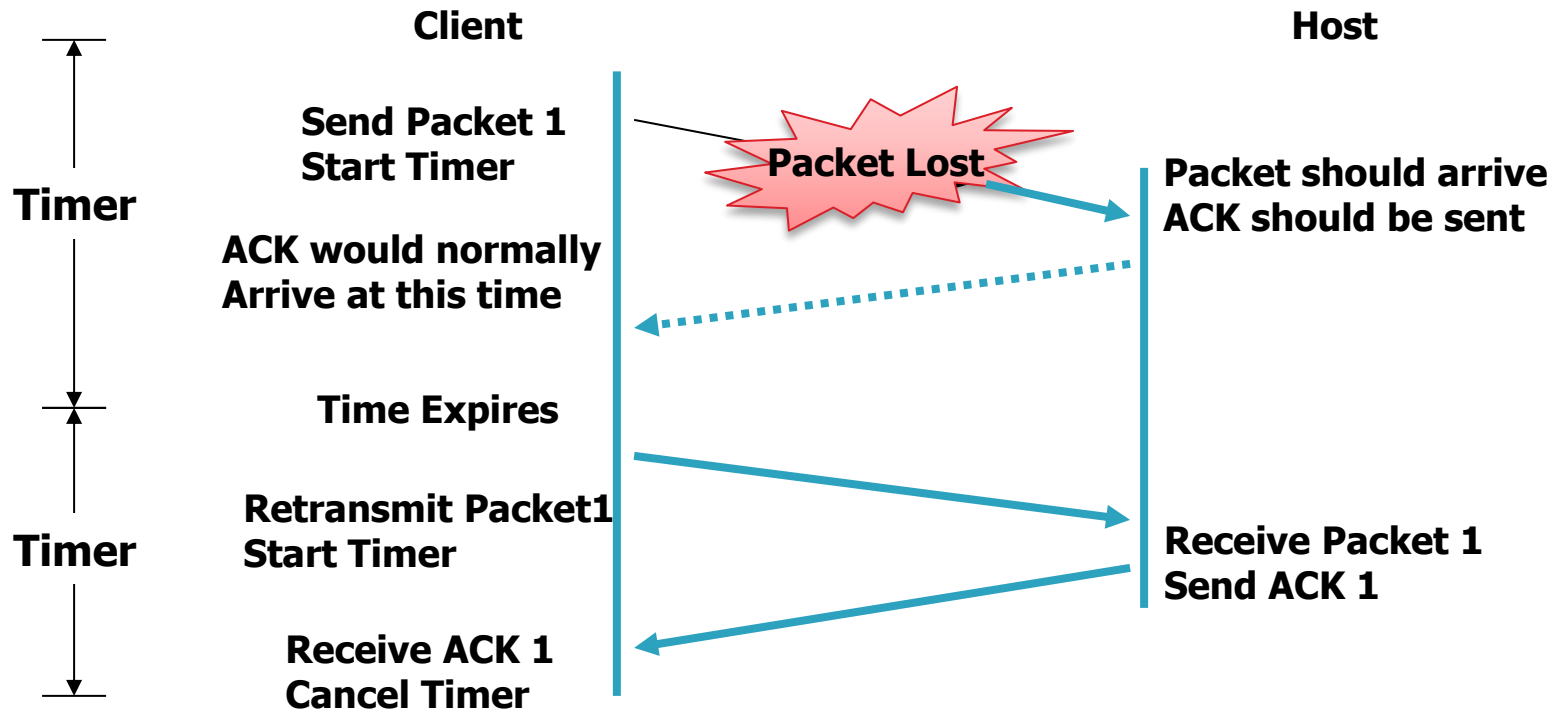
Transmission Control Protocol (TCP)

- ▶ Reliable and the backbone of most of today's protocols based on IP address, which make the communication possible
- ▶ Connections are:
 - Full-duplex
 - Stream-oriented (data can be delivered as stream of bytes)
 - Connection-oriented
- ▶ Protocols that use TCP:
 - HTTP, FTP, SMTP...
- ▶ Saves the application a lot of work, hence mostly used

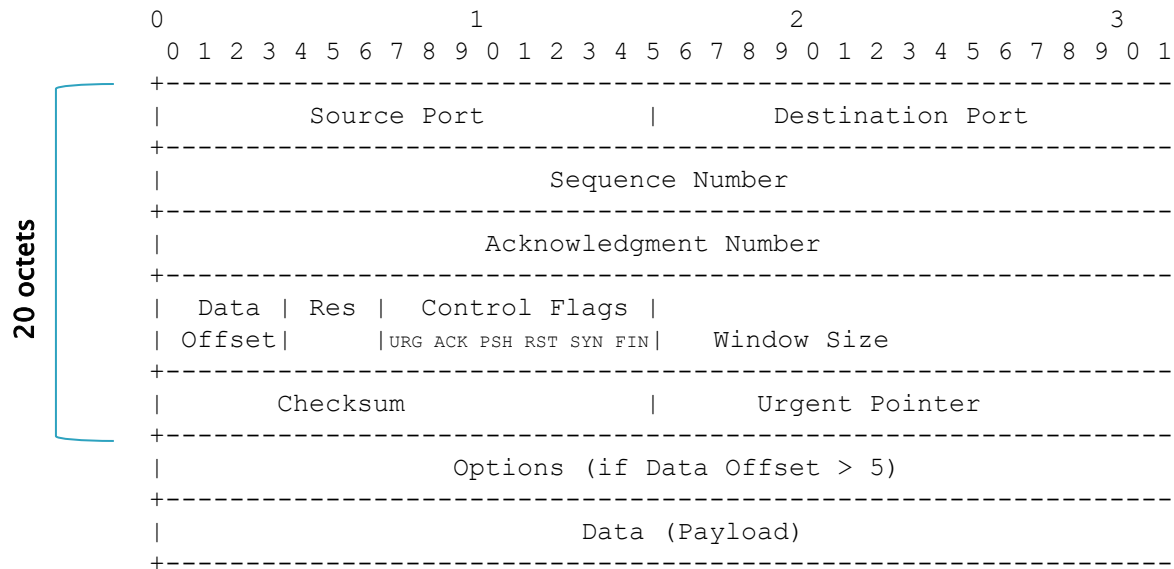
TCP implementation

- ▶ Connections are established using a *three-way handshake* (*Client: Syn, Server: Syn/Ack, Client:Ack*)
- ▶ Data is divided into packets (payload) on OS/application level
- ▶ Packets are sequentially numbered, and received packets are acknowledged
- ▶ Connections are explicitly closed (or may abnormally terminate)

TCP: Data transfer with Timer



TCP Header (1 / 2)



Max. size?
(depends
on the MTU
length)

Field

Source Port

Destination Port

Sequence Number

Acknowledgment #

Flags

Window

Checksum

Urgent Pointer

Options

Payload

Purpose

Identifies originating application

Identifies destination application

Sequence number of first octet in the segment

Sequence number of the next expected octet (if ACK flag set)

TCP flags: [SYN, FIN, RST](connection man.), PSH (push now), ACK (valid), URG (urgent)

Number of octets from ACK that receiver will accept for flow control

Checksum of IP pseudo-header + TCP header + data

Pointer to end of "urgent data" (with URG flag), from the first byte, for priority transfer

Special TCP options such as MSS and Window Scale / variable length

Transmitted data of variable length

TCP Header (2/2)

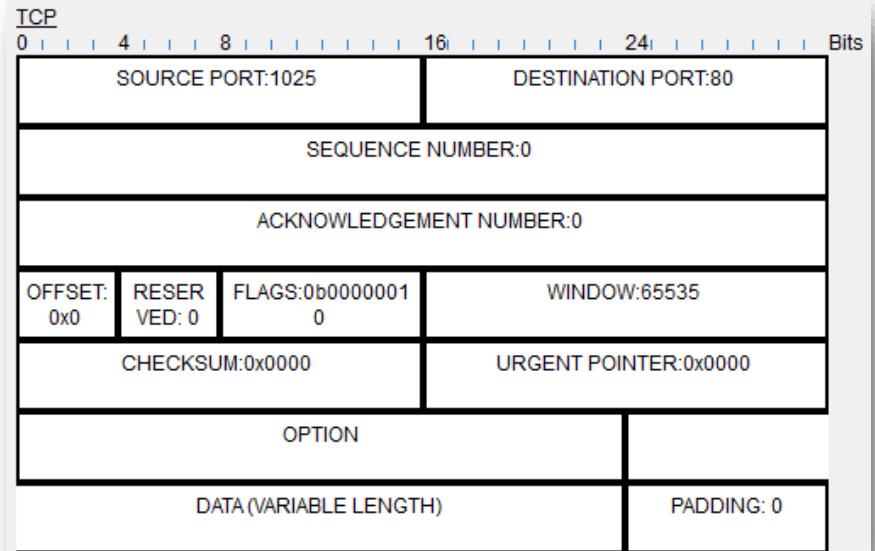
Example from

Wireshark [\(source\)](#)

```
Transmission Control Protocol, Src Port: 443, Dst Port: 50288, Seq: 2066761, Ack: 1, Len: 1412
  Source Port: 443
  Destination Port: 50288
  [Stream index: 2]
  [Conversation completeness: Incomplete (12)]
  [TCP Segment Len: 1412]
  Sequence Number: 2066761 (relative sequence number)
  Sequence Number (raw): 3963077055
  [Next Sequence Number: 2068173 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 3362890618
  0101 .... = Header Length: 20 bytes (5)
  Flags: 0x010 (ACK)
    000. .... = Reserved: Not set
    ...0 .... = Nonce: Not set
    ....0... = Congestion Window Reduced (CWR): Not set
    ....0... = ECN-Echo: Not set
    ....0... = Urgent: Not set
    ....1... = Acknowledgment: Set
    ....0... = Push: Not set
    ....0... = Reset: Not set
    ....0... = Syn: Not set
    ....0... = Fin: Not set
  [TCP Flags: .....A....]
  Window: 245
  [Calculated window size: 245]
  [Window size scaling factor: -1 (unknown)]
  Checksum: 0x2234 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  [Timestamps]
    [Time since first frame in this TCP stream: 6.595641000 seconds]
    [Time since previous frame in this TCP stream: 0.000000000 seconds]
  [SEQ/ACK analysis]
    [Bytes in flight: 4236]
    [Bytes sent since last PSH flag: 25416]
  TCP payload (1412 bytes)
  [Reassembled PDU in frame: 3400]
  TCP segment data (1412 bytes)
```

Example from

Cisco Packet tracer [\(source\)](#)



TCP/IP versions

- ▶ TCP/IP Versions
 - Version 4 (current)
 - Version 5
 - Version 6 (future)
- ▶ IPv6 has been slow to arrive
- ▶ IPv6 requires new software; IT staffs must be retrained; ...
 - Per some experts, IPv6 will most likely coexist with IPv4 for years to come

Lecture 4 Outline

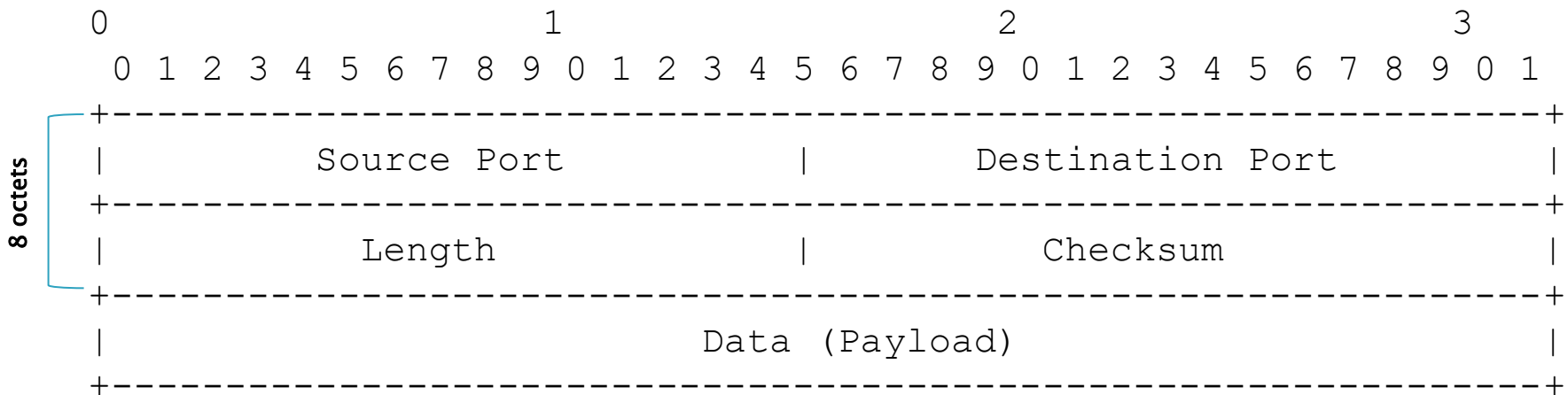
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User Datagram Protocol (UDP)

- ▶ UDP is connectionless, and does not establish any end-to-end connection manager to check on the received packets
 - Packet loss is better handled by application than the network stack (due to no need of unwanted connection setup overhead)
- ▶ It provides port information for application connection
- ▶ Protocols that use UDP:
 - DNS, SNMP, ...

UDP Header (1 / 2)

- ▶ It provides source and destination port numbers for application connection
 - Length: Length field contains the length of the entire UDP segment (UDP header & data)



Max. size?
(depends
on the MTU
length)

UDP Header (2 / 2)

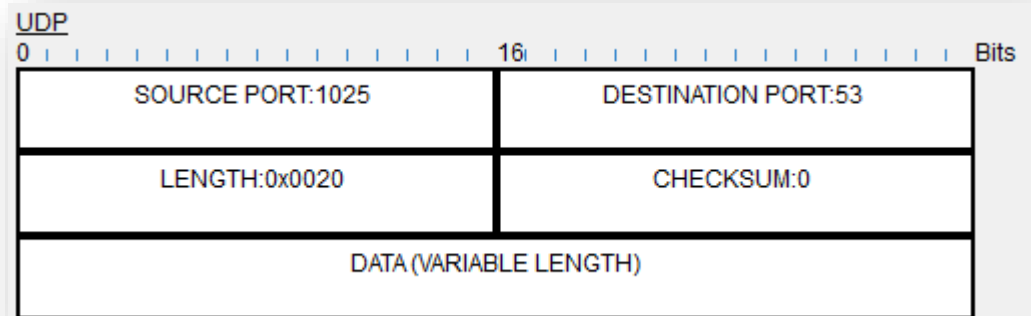
Example from

Wireshark [\(source\)](#)

```
▼ User Datagram Protocol, Src Port: 53, Dst Port: 50180
  Source Port: 53
  Destination Port: 50180
  Length: 174
  Checksum: 0x8786 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 24]
  ▼ [Timestamps]
    [Time since first frame: 0.320422000 seconds]
    [Time since previous frame: 0.275256000 seconds]
  UDP payload (166 bytes)
```

Example from

Cisco Packet tracer [\(source\)](#)



UDP Header (cont.)

Checksum detects bit errors in the UDP header

- It uses the same error checking algorithm as used in TCP and IPv4 headers *i.e. One's complement*

E.g. Bit stream: 11100110011001101101010101010101

1. Divide bit stream into two parts of 16-bit each:

1110011001100110 and 1101010101010101

2. Add these two bit streams: **Binary addition**

1110011001100110

1101010101010101 +

11011101110111011 (wrap around)

1011101110111100 (sum)

3. Apply one's complement (inverse) to the result:

0100010001000011 (checksum)

4. Receiver will add the **sum** and **checksum** value to get all 1's

If an error is detected, the packet is discarded, no error recovery action is taken

Use of the checksum field in UDP is optional (e.g., all bits can be set to zero or off when transmitting data over LAN, since transmission errors can be obtained from Ethernet protocol checksum)

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Internet Control Message Protocol (ICMP)

- ▶ If an intermediary device fails, or if a destination device is disconnected from the network, data cannot be delivered
 - The IP basic design does not notify the sender (reliably) that a data transmission has failed
- ▶ ICMP is a part of TCP/IP protocol stack that was specified in RFC 792 (IPv4)
 - Used to send error messages (not application data) and operational information (indicating success or failure) when communicating with another IP address e.g., IP status, error messages between hosts and routers, ...
 - Uses IP protocol to route its messages between hosts

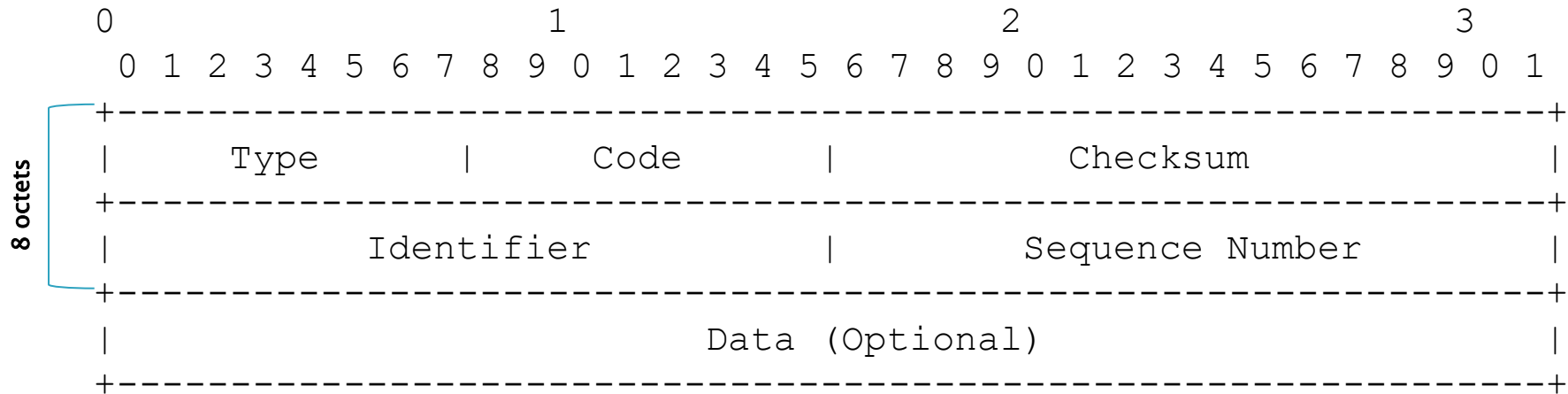
Internet Control Message Protocol (ICMP) (cont.)

- ▶ Needs to be implemented with IP
 - Remember, IP is just a packet delivery system:
 - Transmits and routes datagrams from source to destination through a series of interconnected networks
 - It has a checksum in the IP header to detect lost bits
 - No error detection on the datagram payload though
 - Has no native mechanism for source host notification
 - This is where ICMP comes in:
 - It is used to report IP errors to the source host
- ▶ ICMP Message types:
 - 8: Echo, 0: Echo reply (i.e., PING)
 - 3: Destination unreachable
 - ...

→ Complete reference:

<https://www.iana.org/assignments/icmp-parameters/icmp-parameters.xhtml>

ICMP Header (1 / 2)



- * Not part of ICMP: Can be used arbitrarily by the user
- ▶ Beside reporting delivery errors back to the source of the packet
 - It does not correct the encountered network problem; it merely reports the problem
 - It does not propagate information about network changes to routers
 - The status of the delivered packet gets reported only to the source device

ICMP Header (2 / 2)

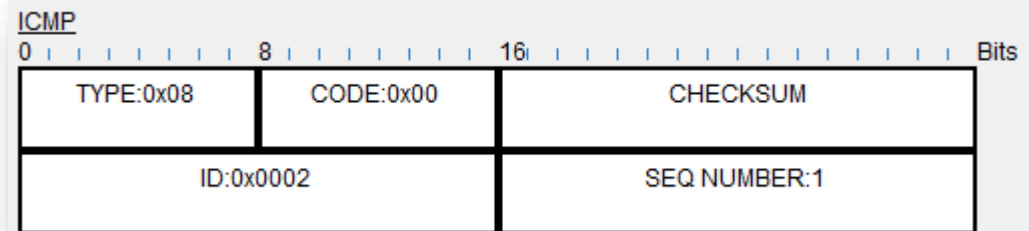
Example from

Wireshark ([source](#))

```
Internet Control Message Protocol
  Type: 0 (Echo (ping) reply)
  Code: 0
  Checksum: 0x5550 [correct]
  [Checksum Status: Good]
  Identifier (BE): 1 (0x0001)
  Identifier (LE): 256 (0x0100)
  Sequence Number (BE): 11 (0x000b)
  Sequence Number (LE): 2816 (0x0b00)
  [Request frame: 21]
  [Response time: 31.140 ms]
  Data (32 bytes)
    Data: 6162636465666768696a6b6c6d6e6f7071727374757677616263646566676869
    [Length: 32]
```

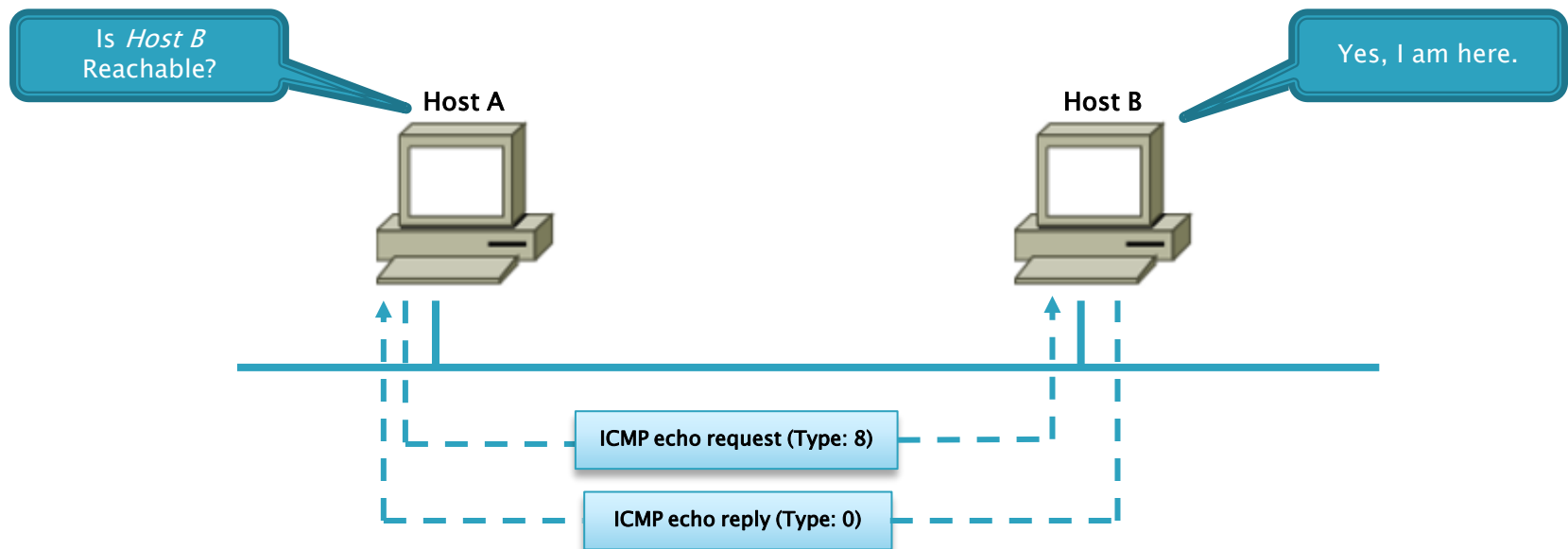
Example from

Cisco Packet tracer ([source](#))



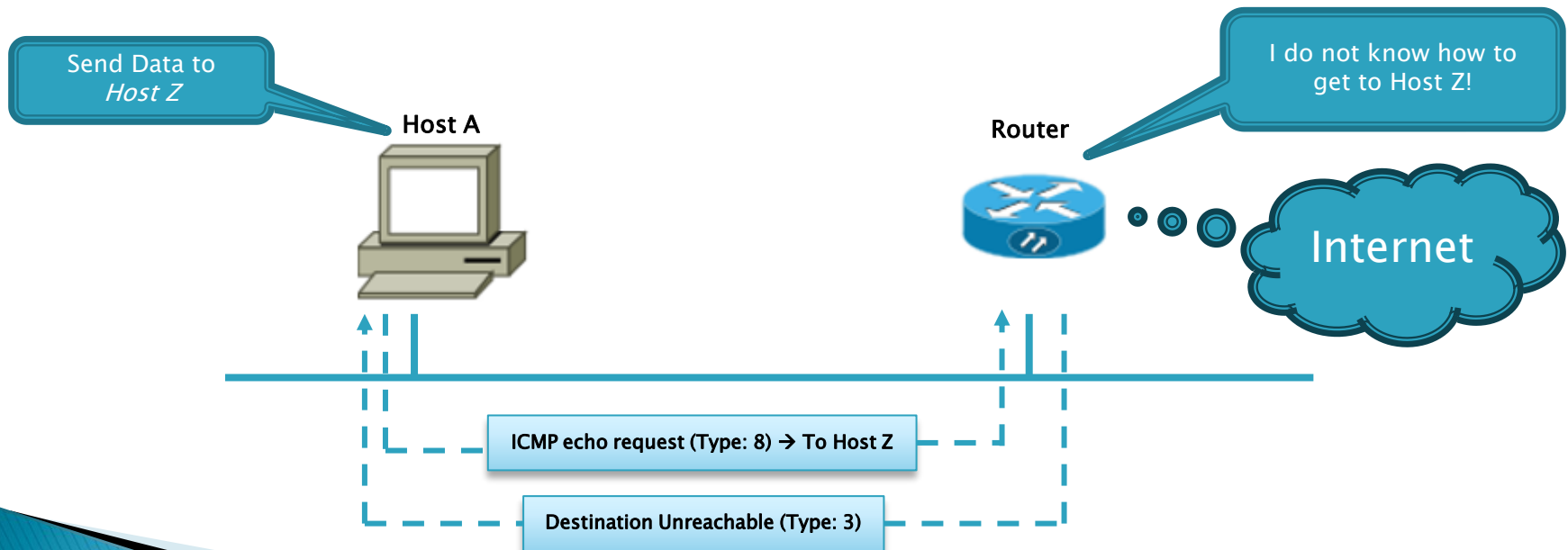
Using ping to test destination reachability (cont.)

- ▶ Traffic generated by the “ping” command



Destination unreachable

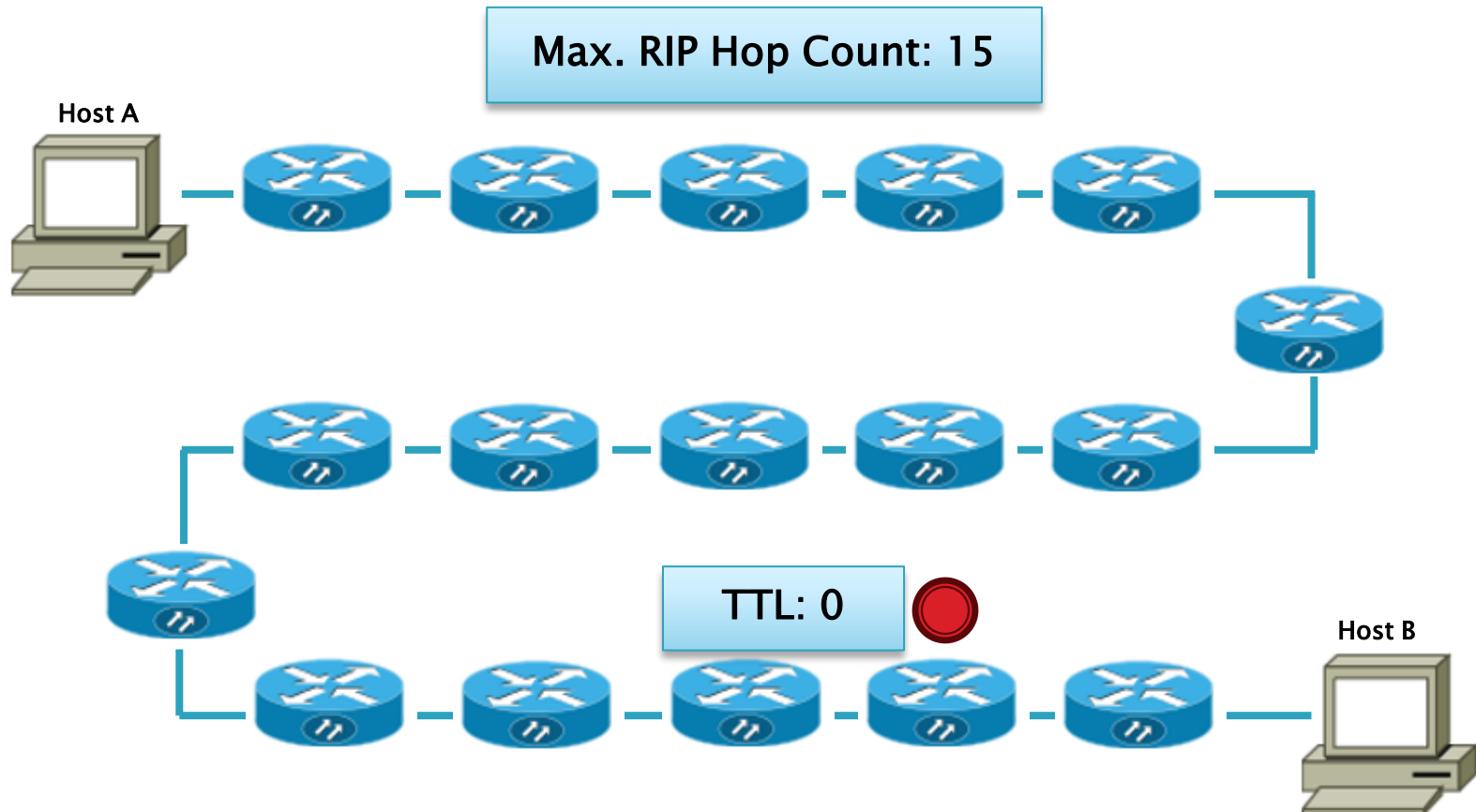
- ▶ An ICMP Destination unreachable message is sent if:
 - Host (code: 1) or port (code: 3) is unreachable
 - Network unreachable (code: 0)



Possible causes

- ▶ For instance, the sending device may address the datagram to a non-existent IP address or to a destination device that is disconnected from its network
- ▶ Routers can also be points of failure e.g., if a connecting interface is down or if the router does not have the information necessary to find the destination network
 - If a destination network is not accessible, it is said to be an unreachable network

Detecting excessively long routes



Detecting excessively long routes (cont.)

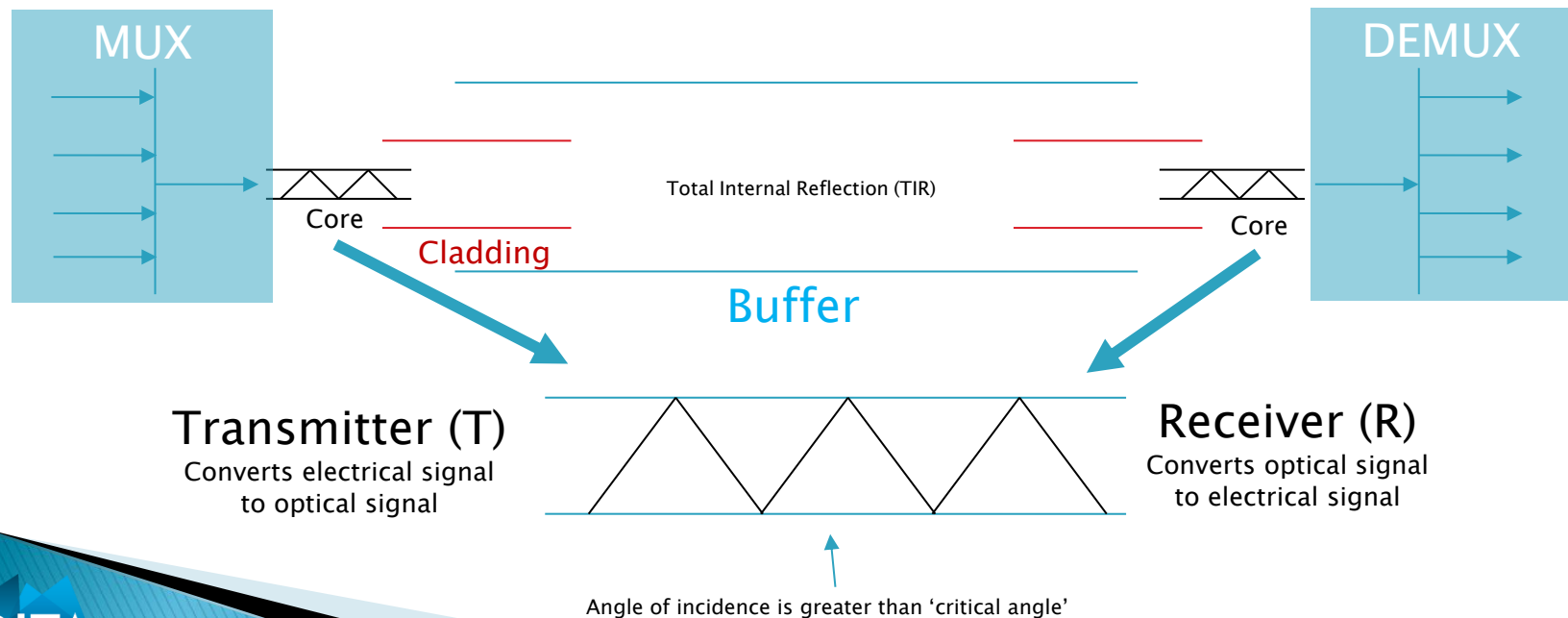
- ▶ Limitations of the routing protocol can result in destinations being unreachable (e.g., Hop 15 count limit in RIP)
- ▶ Whether the actual path includes a circular routing path or too many hops, the packet will eventually exceed the maximum hop count
 - This is also known as reaching its time-to-live (TTL), because the TTL value 'typically' matches the maximum hop count defined by the routing protocol
 - As each router processes the packet, it decreases the TTL value by one, and eventually when the value reaches zero, the packet is discarded
- ▶ ICMP uses a “Type 11 — Time Exceeded” message to notify the source device that the TTL of the packet has been exceeded

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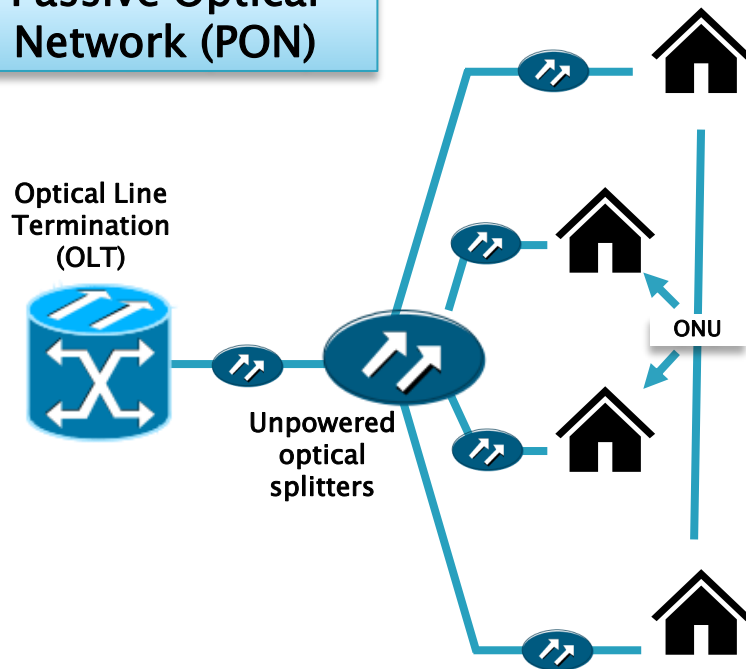
Optical Network (ON)

- ▶ Connects devices using 'optical fiber'
 - Optical fibers are very thin glass cylinders or filaments which carry signals in the form of light
 - Working principal:
both single-mode (9/125 microns), multi-mode (50/125 microns)

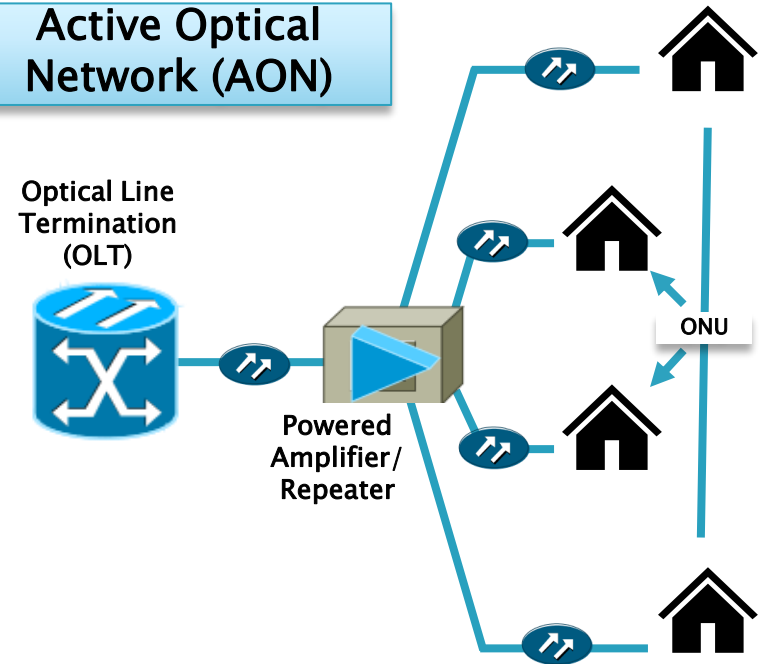


Optical Network Units (ONUs)

Passive Optical Network (PON)



Active Optical Network (AON)



*Point to MultiPoint (P2MP): Shared bandwidth
Distance: ~(10–20)km; residential use (FTTH)*

*Point to Point (P2P): Dedicated bandwidth
Distance: ~(80–100)km; business use
(FTTO, FTTZ)*

Types of Optical Network setups

► Deployment Models:

- **Fiber to the Home (FTTH):** Fiber reaches directly to the user's residence. Most future-proof
- **Fiber to the Premises (FTTP):** Generic term covering both FTTH (home) and FTTB (building)
- **Fiber to the Building (FTTB):** Fiber reaches a building; internal cabling (copper/Ethernet) connects units
- **Fiber to the Curb/Cabinet (FTTC):** Fiber reaches a distribution point near the home (within ~300m); copper (DSL) for last leg
- **Fiber to the Node/Neighborhood (FTTN):** Fiber reaches a node (serving many homes); copper (often VDSL) goes from node to homes
- **Fiber to the Office (FTTO):** Business-focused version of FTTH
- **Fiber to the Zone (FTTZ):** Used in enterprise/campus environments; hybrid setups

Why do we need ON?

- ▶ Growing online users
- ▶ Increasing bandwidth demand
- ▶ Why do we use fiber in ON's?
 - Cost effective with small size & weight
 - Carries large amount of information (upto 10 GBPS)
 - Long distance transmission without regeneration (i.e., less no. of wires & repeaters needed)
 - Electric isolation & signal security (CRC errors)
 - Resistance to High temperature

(i.e., silica melting point 1,713 °C – source: https://en.wikipedia.org/wiki/Silicon_dioxide)

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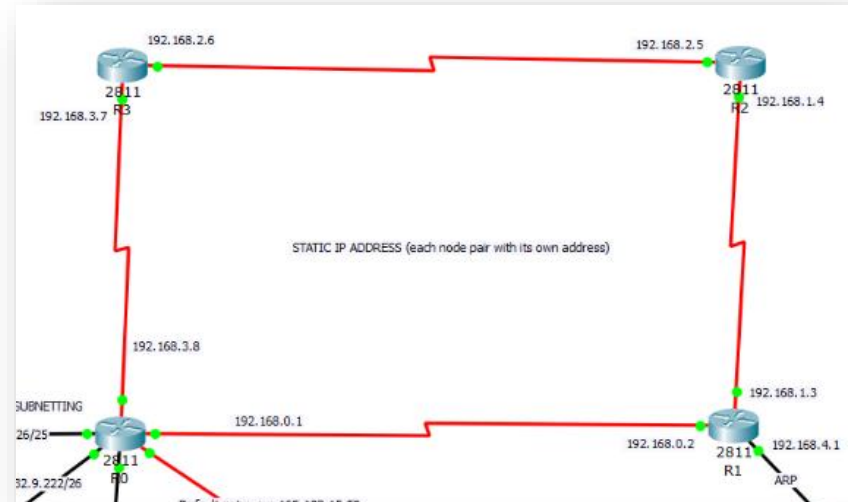
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Exercise 10: Practical work

- ▶ Use your existing cisco packet tracer file (from last class exercise):

1. Simulate network packets transmission using following protocols:
 1. TCP
 2. UDP
 3. ICMP
2. Study each of the protocol packet and document your observations [e.g., What do you see in a specific header field? Does the field value corresponds to a specific octet? If yes, which one is it (e.g., last, first?)] in a separate .doc/.pdf file

It is preferable to use Wireshark



Verify using PING and TRACERT/TRACEROUTE

Lecture 4 ends here

- ▶ **Course Slides: Go to MS Teams:**
Introduction to Network protocols and architecture
→ Files section
- ▶ **Send your questions by email**
OR via direct message using MS Teams
- ▶ **Thank You!**